

Chapter 16 – Wave Motion

Propagation of a disturbance
Speed of waves in strings

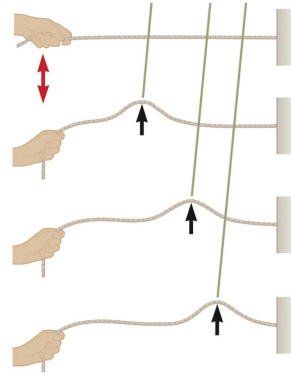
Analysis Model: Traveling Wave
Reflection and Transmission

Waves

- ▶ Central feature to wave motion: **energy is transferred** over a distance, but **matter is not**.
- ▶ Types of waves:
 - **Mechanical waves** – water waves, sound waves and seismic waves. Governed by Newton's laws. Only exist in material mediums.
 - **Electromagnetic waves** – infrared – visible – ultraviolet light, radio and TV waves, microwaves, x-rays and radar waves. Requires no material medium to exist.
 - **Matter waves** – electrons, protons, atoms, molecules and other fundamental particles.

Propagation of a Disturbance

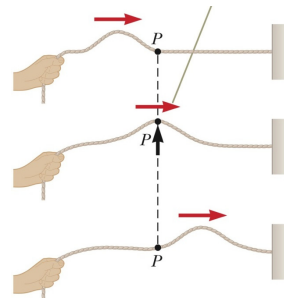
- ▶ **All mechanical waves require:**
 - Some source of disturbance
 - Medium containing elements that can be disturbed
 - A physical mechanism through which elements of the medium can influence each other.
- ▶ Flick one end of a taut string that is fixed at the other end.
- ▶ **Pulse**
 - Hand is the source of disturbance
 - String is the medium
 - Elements of the string are connected.



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Propagation of a Disturbance

- ▶ **Transverse Waves:** Displacement of each particle is perpendicular to direction in which wave is traveling
- ▶ **Longitudinal Waves:** Particles are displaced in the same direction as wave motion.



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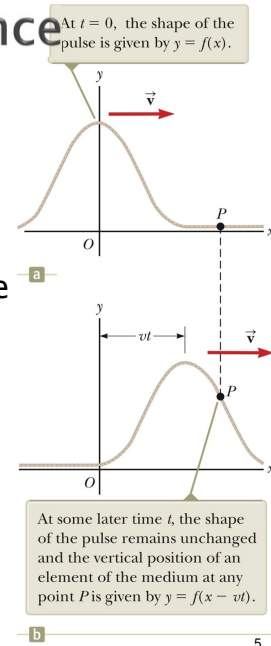
Propagation of a Disturbance

- ▶ Pulse travelling to the right along a string.
- ▶ Shape and position of pulse at $t = 0$
 $y(x, 0) = f(x)$
- ▶ Speed of the pulse v , moved a distance vt at the time t .
- ▶ Shape of pulse does not change.
- ▶ An element of the string at x at time t has the same y position as an element at $x - vt$ at $t = 0$.

$$y(x, t) = y(x - vt, 0)$$

$$y(x, t) = f(x - vt) \text{ traveling to the right}$$

$$y(x, t) = f(x + vt) \text{ traveling to the left}$$

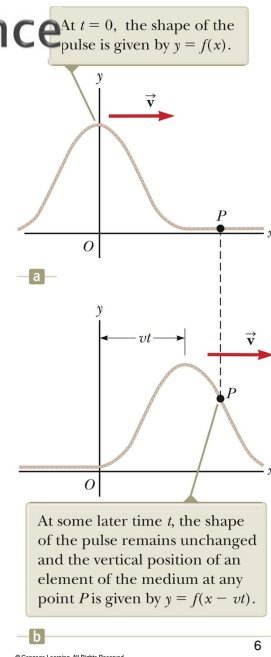


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Propagation of a Disturbance

- ▶ Wave function: $y(x, t)$
- ▶ Wave form: Snapshot at a certain t .
- ▶ Consider particle at point P as time passes. Starts at $y = 0$, increases to max, then returns to zero.



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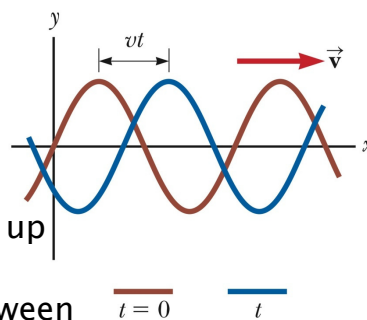
Example 16.1

- ▶ Work through the problem

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Analysis model: Traveling Wave

- ▶ **Sinusoidal Wave:** Can be set up in a string.
- ▶ An important wave function
- ▶ Brown curve: $t = 0$
- ▶ Blue curve: t
- ▶ Wave moves to the right
- ▶ If we focus on an element of the medium – element moves up and down
- ▶ Important to differentiate between the motion of the wave and the motion of the elements of the medium.

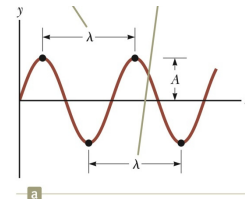


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Analysis model: Traveling Wave

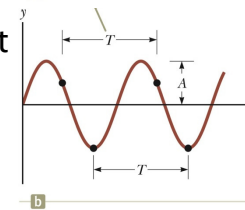
- ▶ Top figure: Traveling wave through medium y vs x

- Crest
- Trough
- Wavelength – λ



- ▶ Bottom figure: position of an element as a function of time y vs t

- Period – T
- Frequency – $f = \frac{1}{T}$
- Amplitude – A



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Analysis model: Traveling Wave

$$y(x,t) = A \sin(kx - \omega t)$$

- ▶ **Amplitude** A : maximum displacement from the position of rest. For transverse wave, transverse displacement y
- ▶ **phase** $\equiv (kx - \omega t)$
- ▶ **Wavelength** λ : distance between repetitions of the wave shape.
- ▶ **Angular wave number**: $k \equiv \frac{2\pi}{\lambda}$ (rad/m)
- ▶ **Period**: time between repetitions of the motion
- ▶ **Angular frequency** (rad/s) $\omega \equiv \frac{2\pi}{T}$
- ▶ **Frequency** (Hz) : $f = \frac{1}{T} = \frac{\omega}{2\pi}$
- ▶ **Wave speed**: $v = \lambda f = \frac{\omega}{k}$

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Analysis model: Traveling Wave

- ▶ General form for a wave function

$$y(x,t) = A \sin(kx - \omega t + \phi)$$

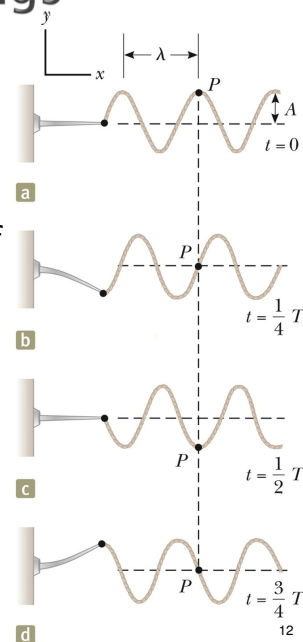
- ▶ ϕ is the phase constant – takes into consideration when the wave does not start at $y = 0$ at $t = 0$.

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Sinusoidal Waves on Strings

- ▶ 1 pulse vs series of pulses
- ▶ A wave
- ▶ SHM of the arm
- ▶ Snapshots of the wave at intervals of $T/4$
- ▶ Each element of string – SHM
- ▶ BUT wave moves to the right $+x$ direction with a speed v .
- ▶ At $t = 0$ – configuration of string (a)

$$y(x,t) = A \sin(kx - \omega t)$$



Sinusoidal Waves on Strings

$$y(x, t) = A \sin(kx - \omega t)$$

- ▶ This expression describes motion of an element on the string

- ▶ Point P x -co-ordinate is constant.

- ▶ Transverse speed (not wave speed):

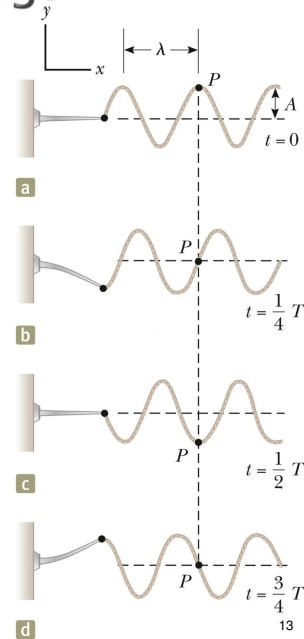
$$v_y = \left. \frac{dy}{dt} \right|_{x=\text{constant}} = \frac{\partial y}{\partial t} = -\omega A \cos(kx - \omega t)$$

- ▶ Transverse acceleration:

$$a_y = \left. \frac{dv_y}{dt} \right|_{x=\text{constant}} = \frac{\partial v_y}{\partial t} = -\omega^2 A \sin(kx - \omega t)$$

- ▶ $v_{y,\text{max}} = \omega A$

$$a_{y,\text{max}} = \omega^2 A$$



Problem

Here are the equations for 3 waves:

$$(1) \quad y(x, t) = 2 \sin(4x - 2t)$$

$$(2) \quad y(x, t) = \sin(3x - 4t)$$

$$(3) \quad y(x, t) = 2 \sin(3x - 3t)$$

Rank the waves according to their (a) wave speed and (b) maximum speed perpendicular to the waves direction of travel (transverse speed).

$$v = \frac{\omega}{k}$$

$$v_{\text{trans}} = \frac{dy}{dt} = \omega y_m \cos(kx - \omega t) \\ = v_m \cos(kx - \omega t)$$

Problem 14

The equation of a transverse wave on a string is

$$y = (2.0 \text{ mm}) \sin\left((20 \text{ m}^{-1})x - (600 \text{ s}^{-1})t\right)$$

What is the amplitude, the frequency, the wavelength and the velocity of the wave?

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Problem

The waves in a microwave oven are described by $\cos(50.3x - 15.1t)$. t measured in ns and x in m.

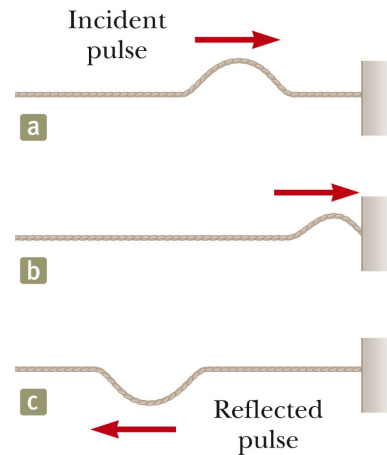
Show that $f = 2.4 \text{ GHz}$, $\lambda = 12.5 \text{ cm}$ and $v = 3.0 \times 10^8 \text{ m/s}$.

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Reflection and Transmission

Reflection : Fixed end

- ▶ When the pulse reaches the support, the pulse moves back along the string in the opposite direction.
- ▶ Reflection of the pulse.
- ▶ The pulse is inverted.
- ▶ Newton's 3rd law
 - When the pulse reaches the fixed end of the string, the string produces an upward force on the support.
 - The support must exert an equal-magnitude and oppositely directed reaction force on the string.

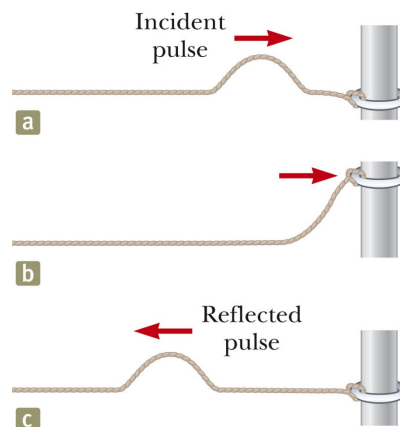


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Reflection and Transmission

Reflection : Free end

- ▶ The string is free to move vertically.
- ▶ The pulse is reflected.
- ▶ The pulse is not inverted.
- ▶ The reflected pulse has the same amplitude as the initial pulse.



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Reflection and Transmission

Transmission of a wave

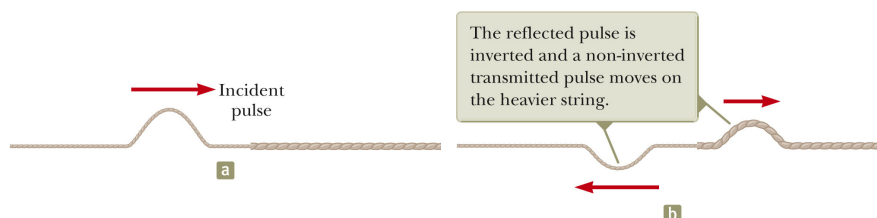
- ▶ When the boundary is intermediate between the last two extremes.
- ▶ Part of the energy in the incident pulse is reflected and part undergoes **transmission**.
- ▶ Some energy passes through the boundary.

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Reflection and Transmission

Transmission of a wave

- ▶ When a light string is attached to a heavier string.

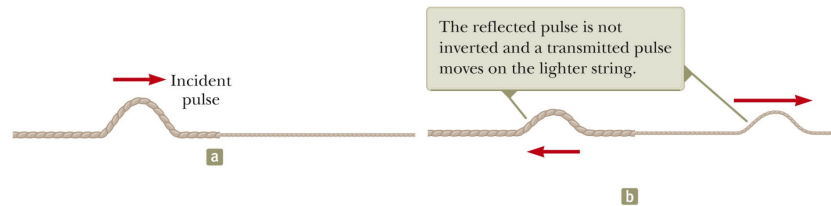


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Reflection and Transmission

Transmission of a wave

- ▶ When a heavy string is attached to a lighter string.



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Chapter 18 – Superposition and standing waves

Waves in interference
Waves under Boundary conditions

Standing waves

Superposition / Interference

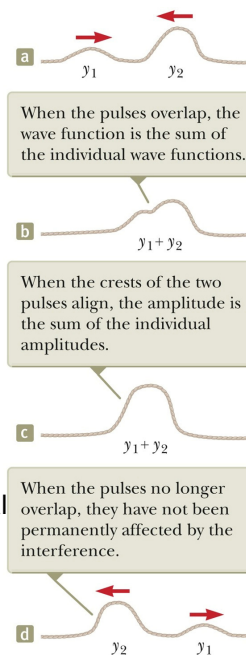


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Superposition

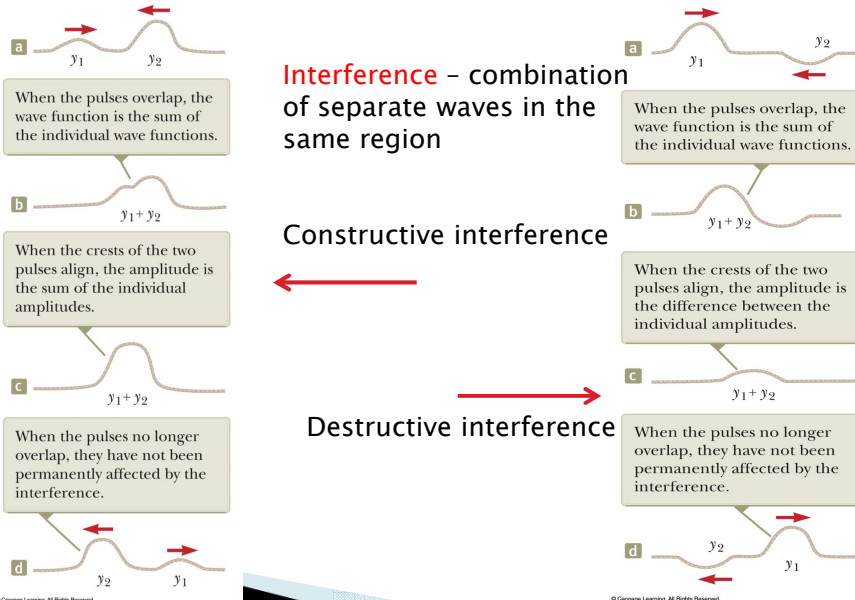
- ▶ Waves vs particles – waves can be at the same position at the same time, not particles.
- ▶ Consider 2 waves travelling along a string with displacements $y_1(x,t)$ and $y_2(x,t)$.
- ▶ When the waves overlap, the displacement is the algebraic sum of the individual displacements:

$$y'_1(x,t) = y_1(x,t) + y_2(x,t).$$
- ▶ **Principle of Superposition** – 2 or more travelling waves moving through a medium, can pass through the same point resulting in a wave function which is the algebraic sum of the individual waves e.g. listening to live music or stereo sound.
- ▶ Overlapping waves do not alter the travel of the waves.



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Superposition



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Superposition of Sinusoidal waves

- ▶ Superposition allows us to add sinusoidal waves of different amplitudes, wavelengths and frequencies.
- ▶ The sum of these waves form an **interference pattern**.
- ▶ Consider a wave: $y_1 = A \sin(kx - \omega t)$
- ▶ And another wave: $y_2 = A \sin(kx - \omega t + \phi)$
- ▶ They have the same ω (and $\therefore f$), the same k (and $\therefore \lambda$) and the same A .
- ▶ They are out of phase by ϕ . (have a phase difference ϕ)

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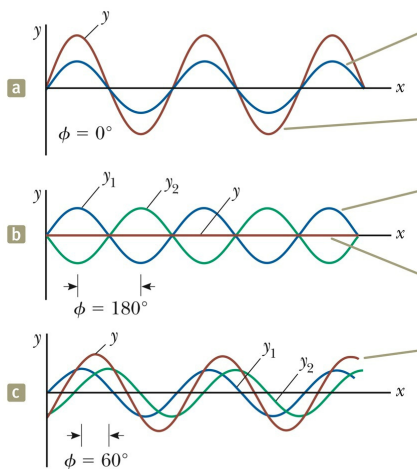
Superposition of Sinusoidal waves

- ▶ $y' = y_1 + y_2$
 $= A \sin(kx - \omega t) + A \sin(kx - \omega t + \phi)$
- ▶ But $\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$
- ▶ Then
 $y' = [2A \cos \frac{1}{2}\phi] \sin(kx - \omega t + \frac{1}{2}\phi)$
- ▶ Resultant
 - Is also sinusoidal
 - has same frequency and wavelength as original individual waves
 - Amplitude: $2A \cos(\phi/2)$
 - If $\phi = 0$, (waves are in phase) then amplitude is $2A$
 - Also when $\phi = 2n\pi$, $n = 0, 2, 4, 6 \dots$

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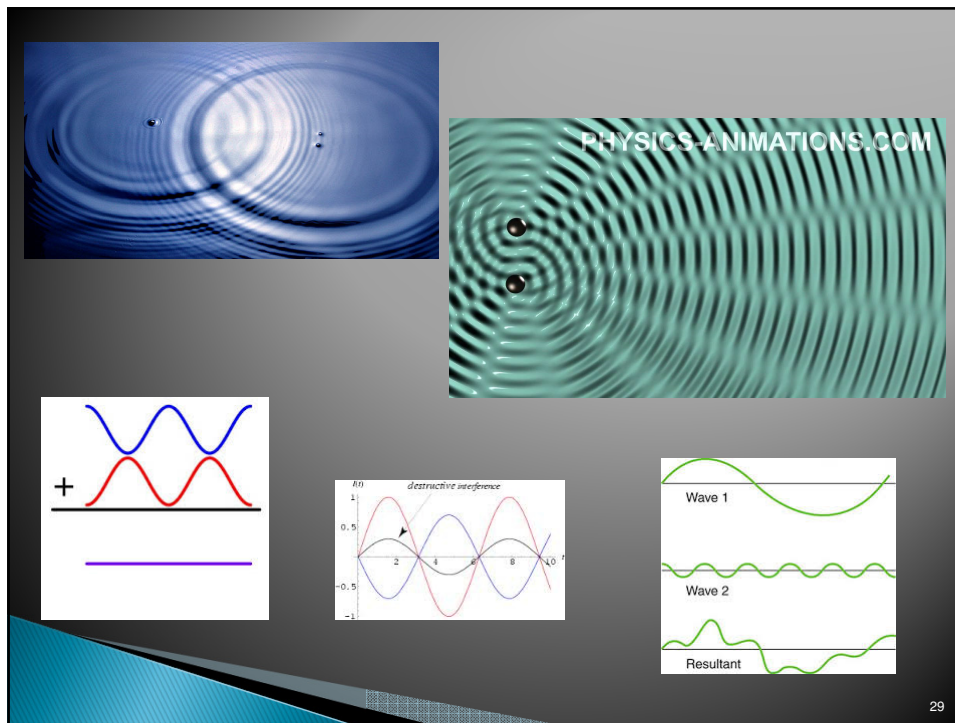
Superposition of Sinusoidal waves

$$y' = [2A \cos \frac{1}{2}\phi] \sin(kx - \omega t + \frac{1}{2}\phi)$$

- 
- ▶ If $\phi = 0$, (waves are in phase) then amplitude is $2A$
 - Also when $\phi = 2n\pi$, $n = 0, 1, 2, 3 \dots$
 - Constructive interference
 - ▶ If $\phi = \pi$, (waves are out of phase) then amplitude is 0
 - Also when $\phi = n\pi$, $n = 1, 3, 5 \dots$
 - Destructive interference
 - ▶ General case: same λ , but different A .
 - If $\phi = 60^\circ$

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Standing waves

- ▶ Previous section - waves travelling in the same direction.
- ▶ This section - waves travel in opposite directions

- ▶ $y_1 = A \sin(kx - \omega t)$ $y_2 = A \sin(kx + \omega t)$

- ▶ Resultant wave function:

$$y'(x,t) = y_1(x,t) + y_2(x,t)$$

$$= A \sin(kx - \omega t) + A \sin(kx + \omega t)$$

- ▶ But $\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$

- ▶ Then $y'(x,t) = \underbrace{(2A \sin kx)}_{\text{amplitude}} \cos \omega t$

- ▶ **Standing waves**

Two sinusoidal waves of same amplitude and wavelength travel in opposite directions along a stretched string, their interference produces a standing wave.



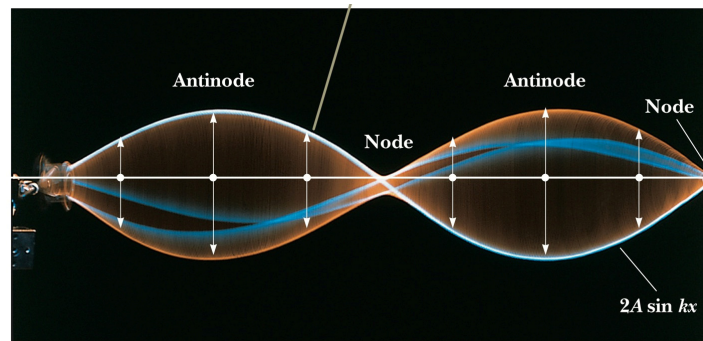
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Standing waves

$$y'(x,t) = (2A \sin kx) \cos \omega t$$

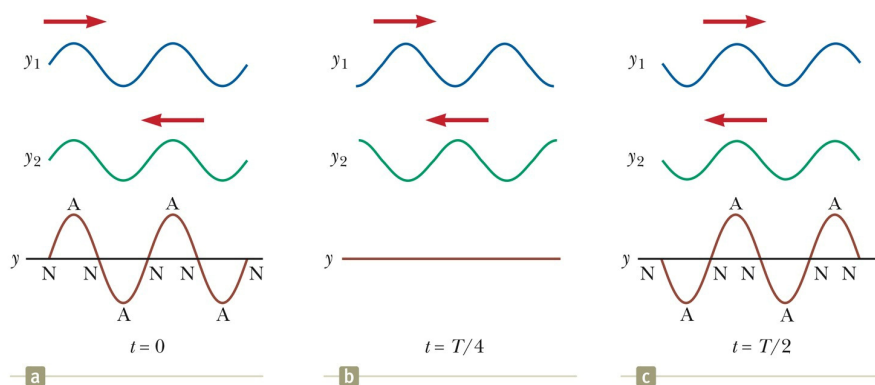
▶ Standing waves

- Two sinusoidal waves of same amplitude and wavelength travel in opposite directions along a stretched string, their interference produces a standing wave.
- There is no $(kx - \omega t)$ in the function – not an expression for a single wave.



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Standing waves



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Standing Waves

- ▶ $y'(x,t) = (2y_m \sin kx) \cos \omega t$
- ▶ **Amplitude zero** where $(\sin kx) = 0$
 $kx = n\pi$ for $n = 0, 1, 2, \dots$
 substitute $k = 2\pi / \lambda$ (wave number)

then $x = n \frac{\lambda}{2}$ For $n = 0, 1, 2, \dots$ (nodes – points of zero amplitude)

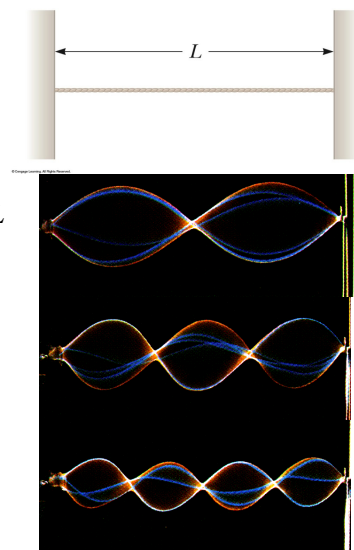
- ▶ **Amplitude maximum** where: $|\sin kx| = 1$
 $kx = (n + \frac{1}{2})\pi$ for $n = 0, 1, 2, \dots$
 substitute $k = 2\pi / \lambda$

then $x = \frac{n\lambda}{4}$ For $n = 1, 3, 5 \dots$ (antinodes – max amplitude)

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Standing waves and Resonance

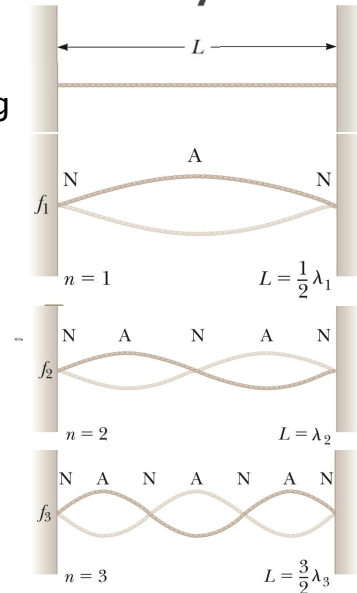
- ▶ Consider a string, stretched between 2 clamps a distance L apart, e.g. guitar.
- ▶ Send a **continuous, sinusoidal wave** toward right.
- ▶ Boundary conditions: $y = 0$ at $x = 0$ and L
- ▶ **Reflects** back and travels toward the left.
- ▶ When reaching the left side, **reflects back...**



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Standing waves and boundary conditions

- ▶ For specific frequencies – standing waves are set up with nodes and anti-nodes.
- ▶ Quantisation
- ▶ Strings will resonate at **resonant frequencies**.
- ▶ Wanneer nie by resonansie f , geen staande golwe nie.

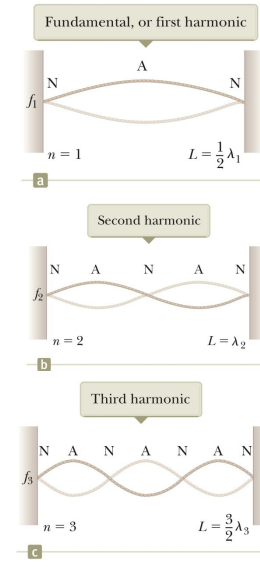


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Standing waves and Resonance

- ▶ Two nodes, one antinode
 $\lambda = 2L$
- ▶ Three nodes, two antinodes:
 $\lambda = L$
- ▶ Four nodes, three antinodes
 $\lambda = \frac{2}{3}L$
- ▶ Relation between λ and L for standing wave

$$\lambda = \frac{2L}{n}, \quad \text{for } n = 1, 2, 3, \dots$$



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Standing waves and Resonance

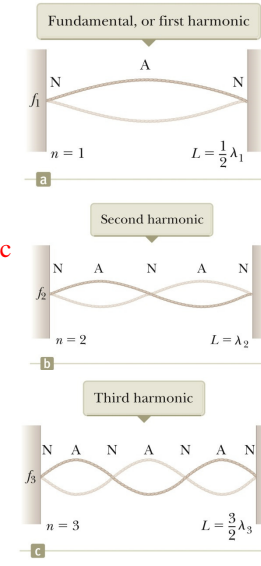
- ▶ In general resonant frequencies:

$$f = \frac{v}{\lambda} = n \frac{v}{2L}, \quad \text{for } n = 1, 2, 3, \dots$$

For $n = 1$: $f_1 = v/2L$ **Fundamental mode or first harmonic**

For $n = 2$: $f_2 = v/L$ **Second harmonic**

For $n = 3$: $f_3 = 3v/2L$ **Third harmonic**



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Problem

- ▶ Standing waves can be excited on a string of length $L = 1.0$ m whose ends are fixed at $x = 0$ and $x = L$. Find the 2 longest wavelengths and 2 smallest values of the wave number k .
- ▶ Wavelength :
- ▶ Wave number:

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Sample Problem 16–8

- ▶ The figure shows a pattern of resonant oscillation of a string of mass $m = 2.500$ g and length $L = 0.800$ m that is under tension $\tau = 325.0$ N. The wave has a velocity of 320 m/s. What is the wavelength λ of the transverse waves producing the standing wave pattern, and what is the harmonic number n ? What is the frequency f of the transverse waves? What is the maximum magnitude of the transverse velocity u_m of the element oscillating at coordinate $x = 0.180$ m. At what point during the element's oscillation is the transverse velocity maximum?

